

Olympiad Test Paper — Science (Class 7)

Chapter 15: Light

Subject	Science (NCERT Class 7)
Chapter	15 — Light
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · –1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. All questions stay within the Class 7 chapter on Light: straight-line travel, reflection, plane and spherical mirrors, lenses, and the dispersion of white light. No calculator is needed. Most questions need two to four reasoning steps; watch for traps — for example, a convex mirror never magnifies, and a virtual image can never be caught on a screen. Do not write on this paper — mark answers on your answer sheet.

1. A torch is switched on in a dusty room and a sharp straight beam is seen. This is the best everyday evidence that light:

- (A) bends around corners
- (B) travels in a straight line
- (C) is made of seven colours
- (D) always slows down in air

2. A small bright lamp is placed in front of a large cardboard sheet that has a tiny hole. The patch of light on a wall behind the hole, and the sharp-edged shadow of the cardboard, both happen because light:

- (A) spreads equally in all directions through solids
- (B) is reflected by the cardboard
- (C) is refracted by the hole
- (D) travels in straight lines

3. An opaque ball is held between a single small lamp and a screen. As the ball is moved **closer to the lamp** (away from the screen), its shadow on the screen will:

- (A) become smaller
- (B) disappear completely
- (C) become larger
- (D) change colour

4. A shadow is always:

- (A) dark and shows only the outline of the object
- (B) coloured like the object
- (C) larger than the object
- (D) the same shape and colour as the object

5. Three opaque cards have holes punched in them. Light from a lamp reaches your eye only when the three holes are:

- (A) of exactly equal size
- (B) placed in a curve
- (C) coloured the same
- (D) arranged in a straight line with your eye and the lamp

6. Light falls on a smooth polished mirror and on a sheet of white paper. Compared with the mirror, the paper:

- (A) absorbs all the light and reflects none
- (B) reflects a sharp clear image like the mirror
- (C) reflects light in many scattered directions (diffuse reflection)
- (D) lets all the light pass straight through

7. You can see a clear image of yourself in a still pond but not in a pond rippled by wind. The reason is that ripples cause:

- (A) refraction of the image
- (B) regular reflection
- (C) diffuse (irregular) reflection
- (D) total absorption of light

8. A ray of light strikes a plane mirror and the angle between the incoming ray and the mirror surface is 30° . The angle of reflection (measured from the normal) is:

- (A) 30°
- (B) 60°
- (C) 90°
- (D) 15°

9. Which statement about the reflection of light from any surface is **always** true?

- (A) Only shiny metals can reflect light.
- (B) Reflection only happens with white light.
- (C) Light is destroyed when it is reflected.
- (D) The angle of incidence equals the angle of reflection.

10. We are able to read the words printed on this paper because the paper produces:

- (A) regular reflection that forms an image
- (B) diffuse reflection that scatters light to our eyes from every part
- (C) its own light
- (D) refraction of the room light

11. The image of your face in a flat bathroom mirror is:

- (A) real and inverted
- (B) real and magnified
- (C) virtual, erect and the same size as you
- (D) virtual and diminished

12. Which one of these can a plane-mirror image **never** be?

- (A) caught on a screen
- (B) erect
- (C) the same size as the object
- (D) laterally inverted

13. You stand 1.5 m in front of a plane mirror. The distance between you and your image is:

- (A) 1.5 m
- (B) 3 m
- (C) 0.75 m
- (D) 0 m

- 14.** A plane mirror forms an image that is the same size as the object. If you walk **towards** the mirror, your image:
- (A) grows larger than you (B) shrinks
(C) turns upside down (D) stays the same size as you
- 15.** A boy stands 2 m from a plane mirror and then steps back to 5 m from it. By how much does his image move away from the mirror?
- (A) 3 m (B) 6 m
(C) 1.5 m (D) 10 m
- 16.** In a plane-mirror image the part that appears swapped (changed) is:
- (A) top and bottom (B) near and far
(C) left and right (D) nothing is swapped
- 17.** Held up to a plane mirror, the printed word **AMBULANCE** would look correct. This is why the word is written reversed on the front of an ambulance, so that:
- (A) a driver ahead reads it correctly in the rear-view mirror
(B) it looks artistic (C) it reflects more light
(D) it can be read in the dark
- 18.** The capital letters H, I, O, X look unchanged in a plane mirror, but the letter R does not. The reason H, I, O, X are unchanged is that:
- (A) they contain only curves (B) they are the brightest letters
(C) they are written in red
(D) they are symmetric about a vertical line down the middle
- 19.** A wall clock shows the time 3:00. Seen in a plane mirror, the reflected clock appears to read:
- (A) 3:00 (B) 6:00
(C) 9:00 (D) 12:00
- 20.** The number **1881** is held up to a plane mirror. Its reflection reads:
- (A) 8118 (B) 1881
(C) 1188 (D) 8811
- 21.** An object is placed 4 cm in front of a plane mirror. The total distance from the object to its image is:
- (A) 4 cm (B) 8 cm
(C) 2 cm (D) 16 cm
- 22.** A candle is placed 10 cm in front of a plane mirror. Where would you have to put a screen to catch the candle's image?
- (A) 10 cm behind the mirror (B) at the surface of the mirror
(C) nowhere — the image cannot be caught on a screen

(D) 10 cm in front of the mirror

23. A man walks towards a plane mirror at 1 m/s. His image approaches the mirror at:

- (A) 1 m/s
- (B) 0.5 m/s
- (C) 2 m/s
- (D) it stays still

24. A reflecting surface curved **inwards** (like the inside of a spoon's bowl) is a:

- (A) plane mirror
- (B) convex mirror
- (C) convex lens
- (D) concave mirror

25. A concave mirror is sometimes called a **converging** mirror because it:

- (A) spreads parallel rays apart
- (B) brings parallel rays together to a point
- (C) lets light pass through it
- (D) splits white light into colours

26. Which mirror can form a **real, inverted** image of a distant object?

- (A) a plane mirror
- (B) a convex mirror
- (C) a concave mirror
- (D) no mirror can

27. When you hold a concave (shaving) mirror **very close** to your face, the image you see is:

- (A) real and inverted
- (B) diminished
- (C) exactly the same size
- (D) virtual, erect and magnified

28. A dentist uses a concave mirror mainly because, held close to a tooth, it gives:

- (A) an enlarged, erect image of the tooth
- (B) a wider view of the whole mouth
- (C) a smaller image so more fits in
- (D) a coloured image

29. Why is a concave mirror, and not a plane or convex one, placed behind the bulb in a torch or car headlight?

- (A) it makes the bulb look bigger
- (B) it spreads the light in all directions
- (C) it changes the colour of the light
- (D) it gathers the light and sends out a strong parallel beam

30. A student claims a concave mirror "always makes things look bigger." This is:

- (A) wrong — a distant object gives a small, inverted image
- (B) correct for every position of the object
- (C) correct only for red objects
- (D) wrong — a concave mirror always shrinks things

31. A reflecting surface curved **outwards** (bulging towards you, like the back of a spoon) is a:

- (A) concave mirror
- (B) plane mirror
- (C) convex mirror
- (D) concave lens

32. Whatever the position of the object, the image formed by a convex mirror is always:

- (A) real and inverted (B) virtual, erect and diminished
(C) virtual, erect and magnified (D) real and the same size

33. A convex mirror is preferred as a vehicle's rear-view mirror chiefly because it:

- (A) gives a wider field of view (B) magnifies cars behind so they look close
(C) forms a real image on the road (D) shows the true distance exactly

34. A salesperson says a security mirror in a shop "magnifies thieves so they look bigger."
For a convex security mirror this claim is:

- (A) correct (B) correct only in bright light
(C) wrong — a convex mirror always gives a diminished image
(D) wrong — it inverts the image instead

35. In a convex mirror, a car that is actually quite close behind looks small and far away.
This happens because the convex mirror always forms a:

- (A) magnified image (B) real image
(C) laterally upright image (D) diminished image

36. Which single property is shared by a plane mirror **and** a convex mirror but **not** by a concave mirror used for a distant object?

- (A) they magnify the object (B) they form a virtual, erect image
(C) they form a real image (D) they invert the image

37. Match the use to the correct mirror: a large mirror used to focus sunlight in a **solar concentrator** is:

- (A) plane (B) a concave lens
(C) convex (D) concave

38. At a blind T-junction a large mirror is fixed high on a wall so drivers can see a **wide stretch** of the crossing road. The right mirror for this is:

- (A) concave, because it magnifies (B) convex, because it gives a wide field of view
(C) plane, because it shows the true size (D) concave, because it gives a real image

39. Four mirrors are described: (i) gives a wide view, (ii) used in a torch reflector, (iii) gives a same-size image, (iv) can form a real image. The pair that are **both concave** is:

- (A) (i) and (iii) (B) (ii) and (iv)
(C) (i) and (ii) (D) (iii) and (iv)

40. A shopkeeper needs a mirror that lets a small camera watch the **whole** shop floor.
The best choice is a:

- (A) convex mirror (B) concave mirror
(C) plane mirror (D) convex lens

41. A lens that is **thicker in the middle** than at the edges is a:

- (A) convex lens
- (B) concave lens
- (C) plane mirror
- (D) convex mirror

42. A magnifying glass used to read tiny print is a:

- (A) concave lens
- (B) concave mirror
- (C) plane mirror
- (D) convex lens

43. A convex lens is called a **converging** lens because rays passing through it are:

- (A) spread apart
- (B) brought together towards a point
- (C) reflected back
- (D) split into colours

44. Which of these can be obtained on a screen (i.e., is a **real** image)?

- (A) the image in a plane mirror
- (B) the image in a convex mirror
- (C) the image of a distant tree formed by a convex lens
- (D) the magnified image seen through a magnifying glass held close to print

45. Sunlight passing through a convex lens can be focused to a tiny bright hot spot on a sheet of paper. This shows that a convex lens:

- (A) spreads light out
- (B) only works with coloured light
- (C) converges light to a point
- (D) reflects sunlight

46. A lens that is **thinner in the middle** than at the edges is a:

- (A) convex lens
- (B) concave mirror
- (C) convex mirror
- (D) concave lens

47. Whatever the position of the object, a concave (diverging) lens always forms an image that is:

- (A) real and inverted
- (B) virtual, erect and diminished
- (C) virtual, erect and magnified
- (D) real and magnified

48. Which one of the following will spread parallel rays of light apart (diverge them)?

- (A) a concave lens
- (B) a convex lens
- (C) a concave mirror
- (D) a magnifying glass

49. A student looks at small print through a concave lens. The print appears:

- (A) larger and upside down
- (B) larger and upright
- (C) smaller and upright
- (D) the same size

50. White light from the Sun is actually a mixture of:

- (A) just three colours
- (B) seven colours
- (C) only red, green and blue
- (D) a single pure colour

51. The correct order of colours in the spectrum of white light, from one end to the other, is given by VIBGYOR, which stands for:

- (A) violet, indigo, blue, green, yellow, orange, red
- (B) red, orange, yellow, green, blue, indigo, violet only if reversed wrongly
- (C) white, indigo, blue, green, yellow, orange, red
- (D) violet, indigo, blue, green, yellow, orange, brown

52. A glass prism held in sunlight throws a band of colours on a wall. The splitting of white light into its colours is called:

- (A) reflection
- (B) dispersion
- (C) lateral inversion
- (D) absorption

53. A rainbow in the sky is formed when sunlight is dispersed by:

- (A) clouds reflecting white light
- (B) tiny water droplets acting like little prisms
- (C) the mirror-like surface of the ground
- (D) a concave mirror in the sky

54. A glass prism splits white light into seven colours. A disc painted with the seven colours and spun very fast (Newton's disc) appears:

- (A) black
- (B) violet
- (C) whitish / white
- (D) striped with all seven colours

55. The fact that a prism can split white light into colours, **and** a spinning seven-colour disc looks white, together show that:

- (A) white light is a single pure colour
- (B) colours are created by the prism itself
- (C) the disc destroys the colours
- (D) white light is made of the seven colours combined

56. A prism produces a band of colours but a flat sheet of window glass does not. The key difference is that the prism's faces are:

- (A) coloured
- (B) made of a special glass
- (C) slanted (not parallel), so the colours stay separated
- (D) curved like a lens

57. If you shine pure **red** light (a single colour) through a glass prism, you will see:

- (A) the full seven-colour spectrum
- (B) white light
- (C) only a band of red light, slightly bent but not split
- (D) no light at all

58. Of the colours in white light, the one usually shown bending the **most** through a prism is at the violet end, and the one bending the **least** is at the:

- (A) red end
- (B) green end
- (C) blue end
- (D) indigo end

59. A teacher shows three demonstrations: (i) a still pond reflecting a tree, (ii) a prism making a rainbow band, (iii) a magnifying glass enlarging print. These illustrate, in order:

- (A) refraction, reflection, dispersion (B) reflection, lateral inversion, dispersion
(C) dispersion, reflection, action of a concave lens
(D) reflection, dispersion, action of a convex lens

60. Which sequence correctly pairs each device with the kind of image it forms?

- (A) plane mirror → virtual same-size; convex mirror → virtual diminished; concave lens → virtual diminished
(B) plane mirror → real; convex mirror → magnified; convex lens (distant object) → virtual
(C) concave lens → real; convex mirror → real; plane mirror → inverted
(D) convex mirror → magnified erect; plane mirror → diminished; concave mirror → never real

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 15: Light

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	C	21	B	31	C	41	A	51	A
2	D	12	A	22	C	32	B	42	D	52	B
3	C	13	B	23	A	33	A	43	B	53	B
4	A	14	D	24	D	34	C	44	C	54	C
5	D	15	A	25	B	35	D	45	C	55	D
6	C	16	C	26	C	36	B	46	D	56	C
7	C	17	A	27	D	37	D	47	B	57	C
8	B	18	D	28	A	38	B	48	A	58	A
9	D	19	C	29	D	39	B	49	C	59	D
10	B	20	B	30	A	40	A	50	B	60	A

Answer-key distribution: A = 14, B = 16, C = 16, D = 14.

Worked Solutions

1. (B) A torch beam appears as a straight line of light because light travels in a straight line (rectilinear propagation). Light does not bend around corners, and the straight beam is not evidence about colour or speed.

2. (D) A patch of light passing through a tiny hole and a sharp-edged shadow both arise because light moves in straight lines: rays that are blocked make the shadow, rays that pass straight through the hole make the bright patch. It is not reflection or refraction here.

3. (C) Moving the opaque ball closer to the small lamp (and farther from the screen) makes it block a wider cone of the diverging rays, so the shadow on the screen grows **larger**. (Moving it closer to the screen would shrink the shadow — the trap.)

4. (A) A shadow is simply the dark region where light is blocked; it shows only the **outline** of the object and has no colour of its own. It is not necessarily larger, and it is never coloured like the object.

5. (D) You see the lamp only when light reaches your eye, which (since light goes in straight lines) requires the three holes, your eye and the lamp to all lie on one straight line. Equal size, colour or a curved arrangement do not matter.

6. (C) A polished mirror reflects rays in one regular direction (regular reflection), giving a clear image. Rough white paper reflects light in many scattered directions — **diffuse (irregular) reflection** — so no image forms, though plenty of light still bounces off.

7. (C) A still water surface acts like a plane mirror (regular reflection) and forms a clear image. Wind makes ripples — an uneven surface — so the reflection becomes scattered, i.e. **diffuse (irregular) reflection**, and the image breaks up.

8. (B) The angle of incidence is measured from the **normal**, not from the surface. If the ray makes 30° with the mirror surface, it makes $90^\circ - 30^\circ = 60^\circ$ with the normal. By the law of reflection the angle of reflection also equals **60°** . (30° is the trap of measuring from the surface.)

9. (D) The fundamental law of reflection is that the angle of incidence equals the angle of reflection — true for every reflecting surface, shiny or dull, and for light of any colour. Reflection is not limited to metals or to white light, and light is not destroyed.

10. (B) Printed paper does not glow on its own; we read it because room light strikes it and undergoes **diffuse reflection**, scattering from every point of the page into our eyes. Regular reflection would give a mirror image, not readable text.

11. (C) A plane mirror always forms a **virtual, erect** image that is the **same size** as the object and as far behind the mirror as the object is in front. It is never real, magnified or diminished.

12. (A) A plane-mirror image is virtual, so it cannot be **caught on a screen**. It is always erect, the same size, and laterally inverted — all of which a plane-mirror image genuinely is. So the one it can never be is "caught on a screen."

13. (B) The image is as far behind the mirror as the object is in front, i.e. 1.5 m behind. The distance from you to your image is therefore $1.5 + 1.5 = \mathbf{3\ m}$. (1.5 m forgets to add the image's distance behind.)

14. (D) A plane mirror always forms an image the same size as the object, regardless of how close you stand. Walking towards the mirror does not change the image's size — it **stays the same size as you**. (It may *appear* to loom, but its true size is unchanged.)

15. (A) The image is always as far behind as the object is in front. At 2 m the image is 2 m behind; at 5 m it is 5 m behind. So the image moves $5 - 2 = \mathbf{3\ m}$ farther back. (6 m is the trap of doubling the object's shift.)

16. (C) A plane mirror swaps **left and right** (lateral inversion); it does not turn the image upside down or swap near and far. Your right hand appears as the image's left hand.

17. (A) In a following driver's rear-view mirror the writing is laterally inverted; printing AMBULANCE already reversed means the mirror reverses it back, so the **driver ahead reads it correctly** and pulls over. It is not for looks, brightness or darkness.

18. (D) Lateral inversion swaps left and right. Letters that are **symmetric about a vertical line** (H, I, O, X) look identical after this swap, while R is not vertically symmetric, so it appears reversed. Brightness or colour is irrelevant.

19. (C) A clock seen in a mirror reads "12-hour minus the real time." For 3:00 the mirror image reads $12:00 - 3:00 = 9:00$. (The hands at the 3 position appear at the 9 position.)

20. (B) Lateral inversion reverses the left-right order and mirror-flips each digit. The digits 1 and 8 are each symmetric, and reversing the order of 1-8-8-1 gives 1-8-8-1 again, so the reflection still reads **1881** (it is a mirror palindrome here).

21. (B) The image lies 4 cm behind the mirror, so object-to-image distance = $4 + 4 = 8$ cm. (4 cm ignores the image distance; 16 cm wrongly squares or doubles twice.)

22. (C) A plane mirror forms a **virtual** image, which is only an apparent point where rays seem to come from — no light actually meets there, so it **cannot be caught on any screen**, whether in front of or behind the mirror.

23. (A) In a plane mirror the image is always as far behind as the object is in front, so the image moves exactly in step with the object. If the man approaches at 1 m/s, his image approaches the mirror at **1 m/s** too. (2 m/s is the rate at which he and his image close on each other, not the image's speed toward the mirror.)

24. (D) A surface that curves **inwards** (the hollow side, like the bowl of a spoon) is a **concave mirror**. The outward-bulging side would be convex.

25. (B) A concave mirror is called converging because incoming parallel rays are reflected so that they **meet (converge) at a point** (the focus). Spreading rays apart describes a convex mirror; passing light through describes a lens.

26. (C) Only a **concave mirror** can form a real, inverted image (of a distant object, the rays actually meet in front of the mirror). A plane mirror and a convex mirror only ever form virtual, erect images.

27. (D) When the object is **very close** to a concave mirror (nearer than its focus), the image is **virtual, erect and magnified** — which is exactly why it is used as a shaving/make-up mirror. Far objects would give a real inverted image, but "very close" gives the enlarged erect one.

28. (A) A dentist holds a concave mirror close to a tooth so it acts like a magnifier, giving an **enlarged, erect image of the tooth** for a better look. A wider view is what a convex

mirror gives, not a concave one.

29. (D) A bulb placed at the focus of a **concave** (converging) reflector sends out a strong, nearly parallel beam — ideal for a torch or headlight. A plane or convex mirror would scatter the light instead of focusing it into a beam.

30. (A) "Always bigger" is **wrong**: a concave mirror magnifies only when the object is close (nearer than the focus). For a **distant object** it gives a small, **inverted** image. So the image depends on object distance.

31. (C) A surface that bulges **outwards** towards you (the back of a spoon) is a **convex mirror**. The inward, hollow side is concave.

32. (B) Whatever the object's position, a convex mirror always forms a **virtual, erect and diminished** image. It can never form a real or magnified image — a key trap.

33. (A) A convex mirror always makes a diminished image, so it packs a **wider field of view** into the glass — letting a driver see more of the road behind. It does not magnify or show true distance; objects look smaller and farther than they really are.

34. (C) "Magnifies thieves" is **wrong**: a convex (security) mirror always gives a **diminished** image. Its value is the wide field of view, letting one mirror cover a large area, not enlargement.

35. (D) A convex mirror always forms a **diminished** image, so a nearby car looks small and far. (This is why such mirrors carry the warning "objects are closer than they appear.")

36. (B) A plane mirror and a convex mirror both form **virtual, erect** images, and neither magnifies. A concave mirror used for a distant object forms a real, inverted image instead. So the shared property is "they form a virtual, erect image."

37. (D) A solar concentrator uses a large **concave** mirror to converge (focus) sunlight onto one spot to produce heat. A convex mirror or a lens would not focus the light to a hot point the way a concave reflector does.

38. (B) A blind-junction safety mirror must show a **wide stretch** of road, so a **convex** mirror is used — it gives a wide field of view. A concave mirror would magnify but show only a narrow region.

39. (B) (i) wide view = convex; (ii) torch reflector = concave; (iii) same-size image = plane; (iv) can form a real image = concave. The two **concave** ones are **(ii) and (iv)**.

40. (A) Wait — re-reading: a single mirror that lets a camera watch the **whole** shop floor needs the widest possible field of view, so a **convex mirror** is the right choice. A concave or plane mirror would not capture the whole floor in one view.

- 41. (A)** A lens that is **thicker in the middle** than at its edges is a **convex lens** (a converging lens). The concave lens is the one thinner in the middle.
- 42. (D)** A magnifying glass is a **convex lens**: held close to small print it gives an enlarged, erect (virtual) image. A concave lens would shrink the print; mirrors are not used as hand magnifiers here.
- 43. (B)** A convex lens is called converging because rays passing through it are **brought together towards a point** (the focus). Spreading rays apart is what a concave lens does.
- 44. (C)** A **real** image can be caught on a screen. Of the four, only the **image of a distant tree formed by a convex lens** is real (the rays actually meet and can fall on a screen). Plane-mirror, convex-mirror and close-up magnifier images are all virtual.
- 45. (C)** Focusing sunlight to a tiny hot spot shows the convex lens **converges light to a point** (its focus). A concave lens would spread the light out instead, and the effect works with ordinary white sunlight.
- 46. (D)** A lens that is **thinner in the middle** than at its edges is a **concave lens** (a diverging lens). The convex lens is thicker in the middle.
- 47. (B)** Whatever the object's position, a concave (diverging) lens always forms a **virtual, erect and diminished** image. It can never form a real or magnified image — the lens analogue of the convex-mirror rule.
- 48. (A)** A **concave lens** is the diverging element: it spreads parallel rays apart. A convex lens and a concave mirror both converge light, and a magnifying glass is a convex lens.
- 49. (C)** A concave lens always gives a virtual, erect, **diminished** image, so the print looks **smaller and upright**. (A convex lens would make it larger — that is the trap.)
- 50. (B)** White sunlight is actually a mixture of **seven colours** (the VIBGYOR spectrum). It is not a single pure colour or merely three colours.
- 51. (A)** VIBGYOR names the colours in order: **violet, indigo, blue, green, yellow, orange, red**. The other options either end in "brown," start with "white," or are deliberately garbled.
- 52. (B)** The splitting of white light into its component colours by a prism is called **dispersion**. Reflection is bouncing back; lateral inversion is the left-right swap in a mirror; absorption is light being taken up.
- 53. (B)** A rainbow forms when sunlight enters countless **tiny water droplets**, each acting like a small prism that disperses the light into its colours. Clouds, the ground, or a sky-mirror do not produce a rainbow.

54. (C) A prism splits white light into seven colours; conversely, a fast-spinning seven-colour (Newton's) disc recombines them, so the eye blends the colours and the disc looks **whitish / white**. This is the reverse of dispersion.

55. (D) A prism separating white light into seven colours, plus a spinning seven-colour disc looking white, together prove that **white light is made of the seven colours combined**. The prism reveals the colours; it does not create them.

56. (C) A prism's two refracting faces are **slanted (not parallel)**, so the colours bent by the first face keep spreading and stay separated. A flat sheet of window glass has parallel faces, so any spreading is undone and no colour band appears.

57. (C) A single pure colour like **red** cannot be split further, so a prism only **bends a band of red light slightly without splitting it** into other colours. Only white (mixed) light gives the full spectrum.

58. (A) In dispersion, violet (shortest path-bending the most) is bent most and **red** is bent the least. So the colour bending the least is at the **red end** of the spectrum.

59. (D) (i) a still pond imaging a tree = **reflection**; (ii) a prism making a colour band = **dispersion**; (iii) a magnifying glass enlarging print = the action of a **convex lens**. In order: reflection, dispersion, action of a convex lens.

60. (A) Correct pairing: a plane mirror gives a **virtual, same-size** image; a convex mirror gives a **virtual, diminished** image; a concave lens gives a **virtual, diminished** image. The other options wrongly claim real/magnified/inverted images for these devices.